IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-082

1. Topic (12 words max)

AI-Powered Intelligent Talent Acquisition System

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The rapid growth of digital recruitment platforms and Applicant Tracking Systems (ATS) has significantly transformed hiring processes; however, most existing systems remain heavily reliant on rule-based filtering, keyword matching, and manual decision-making. These limitations lead to several critical issues, including inaccurate job–candidate matching, shallow skill evaluation, lack of transparency in AI-driven decisions, and an inability to predict long-term hiring outcomes such as early employee attrition. As a result, organizations continue to face high recruitment costs, reduced hiring quality, and early workforce turnover despite adopting automated tools. Current job–candidate matching mechanisms often fail to explain *why* a candidate is recommended, reducing recruiter trust in AI systems. Similarly, resume parsing solutions primarily focus on keyword extraction, ignoring contextual indicators such as experience duration, project complexity, and real-world skill application. This results in an incomplete and sometimes misleading assessment of candidate proficiency. Furthermore, traditional ATS platforms do not address hiring risks proactively, particularly the likelihood of early attrition caused by poor role alignment, unrealistic salary expectations, or unstable career patterns. Another major gap exists in interview evaluation, where assessments are largely subjective and dependent on interviewer judgment. Existing systems rarely analyze communication quality, emotional cues, or behavioral indicators that strongly influence job performance. This creates inconsistency and bias in candidate evaluation.

This research addresses these gaps by proposing an AI-driven intelligent recruitment framework that integrates explainable job–candidate matching, context-aware resume skill proficiency analysis, early attrition risk prediction, and AI-based interview soft-skill evaluation. By leveraging

machine learning, natural language processing, and explainable AI techniques, the proposed system aims to improve decision accuracy, transparency, and long-term hiring success. The research contributes a holistic, data-driven recruitment approach that extends beyond conventional ATS capabilities and supports informed, fair, and sustainable hiring decisions.

6. Brief description of Existing Research and Systems (300 words max)

Existing research and commercial recruitment systems increasingly utilize automation and artificial intelligence to support hiring decisions. Most Applicant Tracking Systems (ATS) provide basic functionalities such as resume parsing, keyword-based job–candidate matching, and applicant ranking. These systems commonly rely on rule-based logic or traditional machine learning techniques that compare predefined skill keywords with job descriptions. While effective for handling large applicant volumes, such approaches often fail to capture contextual meaning, real skill proficiency, or candidate suitability beyond surface-level attributes.

Recent academic research has explored the use of Natural Language Processing (NLP) techniques, including word embeddings and transformer-based models, for resume parsing and job matching. These studies demonstrate improved semantic understanding of resumes and job descriptions compared to keyword-based methods. However, many implementations still focus on skill extraction accuracy rather than evaluating the depth of experience, project relevance, or duration-based skill proficiency. As a result, nuanced candidate capabilities are often overlooked.

Explainable AI (XAI) has gained attention in recruitment research to address transparency concerns. Techniques such as feature importance analysis, attention visualization, and local explanation models have been proposed to justify AI-driven recommendations. Despite these advancements, most existing systems provide limited or generic explanations that are difficult for recruiters to interpret and rarely integrated into production ATS platforms.

Research on employee attrition prediction primarily targets post-hiring scenarios, using organizational data such as performance metrics, attendance, and engagement indicators. Pre-hiring attrition risk assessment remains underexplored, leaving organizations unable to identify high-risk candidates during recruitment. Similarly, AI-based interview analysis research focuses mainly on speech-to-text transcription or sentiment detection, with limited integration of behavioral, emotional, and communication-based soft-skill assessment.

Overall, existing systems address isolated recruitment tasks but lack an integrated, explainable, and predictive framework capable of supporting end-to-end intelligent hiring decisions.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an AI-driven intelligent recruitment framework designed to support transparent, accurate, and data-driven hiring decisions across the entire recruitment lifecycle. The system integrates four core AI modules—resume intelligence, job–candidate matching, interview analysis, and early attrition risk prediction—into a unified architecture that extends the capabilities of traditional Applicant Tracking Systems (ATS).

At the data ingestion layer, the system collects structured and unstructured data from resumes, job descriptions, interview transcripts (text/audio), and candidate metadata such as career history and salary expectations. Natural Language Processing (NLP) techniques are used to preprocess textual inputs through tokenization, normalization, embedding generation, and semantic representation.

The Intelligent Resume Parsing & Skill Proficiency Analysis module extracts candidate skills, education, and experience while estimating skill depth using contextual indicators such as experience duration, project descriptions, and role relevance. This enables a more realistic assessment of candidate capability compared to keyword-based extraction.

The Job–Candidate Matching & Explainable AI Engine computes similarity scores between candidates and job roles using machine learning–based embedding similarity and weighted feature alignment. Explainable AI techniques are integrated to generate human-readable justifications, highlighting matched skills, experience relevance, and missing requirements, thereby increasing recruiter trust and interpretability.

The AI-Based Interview Evaluation & Soft-Skill Analysis module analyzes interview responses to assess communication quality, sentiment, emotional cues, and behavioral indicators. This module transforms subjective interview evaluations into structured insights, supporting fair and consistent candidate assessment.

The AI-Driven Hiring Risk & Early Attrition Prediction module estimates the likelihood of early employee turnover before hiring. It leverages candidate career patterns, job-switching frequency, skill-role alignment, interview expectations, and compensation mismatch signals to identify high-risk hires. This proactive risk assessment enables organizations to reduce recruitment losses and improve workforce stability.

All module outputs are consolidated within a decision-support layer that provides recruiters with ranked candidate recommendations, explainable insights, interview evaluations, and attrition risk scores through an interactive dashboard. The solution emphasizes modularity, transparency, and scalability, making it adaptable to real-world recruitment environments.

System Perspective Input

The system accepts diverse recruitment-related data sources, including:

- Unstructured resumes in multiple formats (PDF, DOCX).
- Job descriptions containing required skills, qualifications, and experience.
- Candidate profile data and application metadata.
- Historical hiring and recruitment outcome data.

Processing Architecture

The processing architecture of the proposed AI-driven recruitment system follows a modular, multi-layered design that enables scalable, explainable, and data-driven hiring analysis. The architecture transforms raw recruitment data into actionable insights through sequential preprocessing, intelligent modeling, and decision-support stages.

At the initial stage, data ingestion and preprocessing are performed on all incoming inputs, including resumes, job descriptions, candidate profiles, and historical recruitment data. Natural Language Processing (NLP) techniques such as text normalization, tokenization, stop-word removal, and semantic embedding generation are applied to unstructured textual data. Structured data is cleaned, encoded, and standardized to ensure consistency across modules.

The Intelligent Resume Parsing and Skill Proficiency Analysis module processes resume content to extract skills, education, experience, and project information. Context-aware feature engineering is applied by considering experience duration, role relevance, and project complexity to estimate actual skill proficiency rather than relying solely on keyword presence.

The Job–Candidate Matching and Explainable AI Engine computes similarity scores between candidate profiles and job descriptions using machine learning–based semantic similarity models and weighted feature alignment. Explainable AI techniques, such as feature contribution analysis, are integrated to generate transparent explanations for each recommendation.

The AI-Based Interview Evaluation and Soft-Skill Analysis module processes interview transcripts or audio-derived text to evaluate communication patterns, sentiment, emotional cues, and behavioral indicators. These outputs are converted into structured soft-skill scores to support objective interview assessment.

The AI-Driven Hiring Risk and Early Attrition Prediction module aggregates outputs from all previous modules along with candidate career history and expectation data to predict the probability of early employee attrition using supervised machine learning models.

Finally, the decision-support layer consolidates all analytical outputs into an integrated recruiter dashboard, providing ranked candidate recommendations, explainable insights, interview evaluations, and attrition risk indicators. This layered processing architecture ensures transparency, modularity, and effective end-to-end recruitment intelligence.

Output

The proposed AI-driven recruitment system generates multiple structured and interpretable outputs to support informed and transparent hiring decisions. These outputs are produced by integrating insights from all system modules and are presented through a decision-support interface for recruiters and hiring managers.

• Job–Candidate Match Scores and Rankings:
Numerical similarity scores representing the degree of alignment between candidate profiles and job requirements. Candidates are ranked based on overall suitability for a given role.

• Explainable Match Insights:
Human-readable explanations highlighting matched skills, experience relevance, missing qualifications, and feature contributions used in job–candidate recommendations. These insights enhance transparency and recruiter trust.

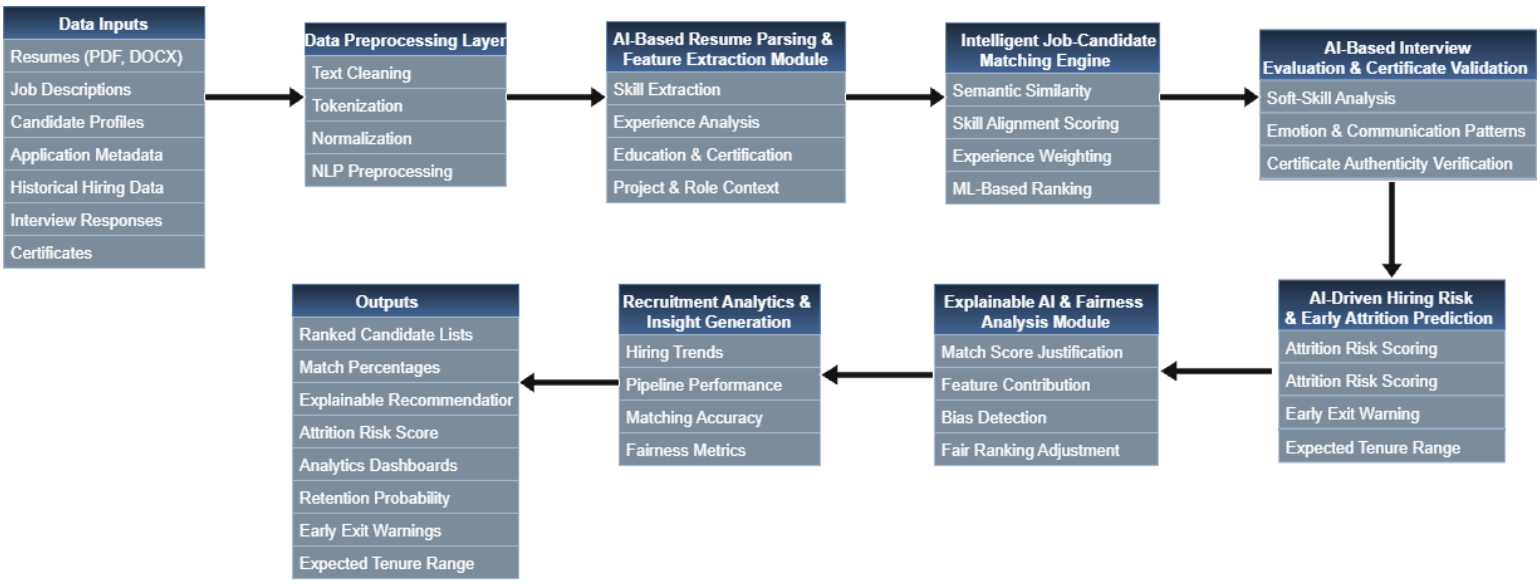
• Skill Proficiency Profiles:
Context-aware evaluations of candidate skills, indicating estimated proficiency levels based on experience duration, project involvement, and role relevance rather than simple keyword presence.

• Interview Evaluation and Soft-Skill Scores:
Structured assessments of communication quality, sentiment, emotional indicators, and behavioral traits derived from interview data. These outputs provide objective support for interview-based decision-making.

• Early Attrition Risk Scores:
Predictive risk indicators estimating the likelihood of early employee turnover prior to hiring. These scores help organizations identify high-risk candidates and reduce recruitment and training losses.

• Recruitment Decision Support Dashboard:
An integrated visualization interface presenting all analytical outputs, enabling recruiters to compare candidates, review explanations, and make data-driven hiring decisions efficiently.

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful development of the proposed AI-driven intelligent recruitment system is supported by the availability of relevant domain expertise, technical knowledge, and accessible data sources. The research team possesses foundational and applied knowledge in software engineering, artificial intelligence, machine learning, and data science, enabling effective system design, implementation, and evaluation. Academic exposure to machine learning algorithms, natural language processing (NLP), explainable AI (XAI), and data analytics provides the theoretical basis required to address complex recruitment challenges.

From a domain perspective, recruitment and human resource management processes are well-documented in both academic literature and industry practices. Existing research on talent acquisition, resume screening, interview evaluation, and employee attrition provides valuable insights into key decision factors, performance indicators, and evaluation metrics. This established body of knowledge enables the research team to model recruitment workflows, define meaningful features, and align system outputs with real-world hiring requirements.

In terms of technical expertise, the team is equipped with experience in implementing NLP pipelines for text extraction, semantic embedding, and contextual analysis, which are essential for resume parsing and job–candidate matching. Knowledge of supervised and unsupervised machine learning techniques supports the development of similarity scoring, soft-skill assessment, and attrition risk prediction models. Additionally, familiarity with explainable AI methods allows the integration of transparent decision explanations, addressing trust and interpretability concerns in AI-driven hiring systems.

Data availability is another critical factor supporting the feasibility of the research. Publicly available datasets such as anonymized resumes, job descriptions, interview transcripts, and employee attrition records can be leveraged for model training and validation. Where real-world recruitment data is limited, synthetic data generation and data augmentation techniques can be

applied to simulate realistic hiring scenarios while preserving ethical and privacy considerations. Historical recruitment outcome data, even in limited quantities, is sufficient to demonstrate proof-of-concept predictive modeling within an academic research context.

Furthermore, modern machine learning frameworks, open-source NLP libraries, and cloud-based development tools are readily accessible, reducing implementation barriers and supporting scalable experimentation. Ethical guidelines, fairness frameworks, and data privacy standards are also well-established, enabling responsible handling of sensitive recruitment data.

Overall, the availability of domain expertise, technical knowledge, and accessible data resources provides a strong foundation for developing, validating, and evaluating the proposed intelligent recruitment system within the scope of an undergraduate final-year research project.

9. Objectives and Novelty

Main Objective

To design and develop an AI-powered intelligent talent acquisition system that automates resume screening, enables accurate and explainable job–candidate matching, reduces bias, and provides data-driven recruitment insights to support fair and efficient hiring decisions.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Fonseka WGK IT22248794	Job–Candidate Matching & Explainable AI Engine	Develop a machine learning–based matching engine to compute job–candidate similarity scores. Implement explainable AI mechanisms to justify recommendations.	Provides transparent and interpretable match explanations, increasing recruiter trust and decision-making clarity.
Jayantha CH IT22226914	Intelligent Resume Parsing & Skill Proficiency Analysis	Design and implement an NLP-based resume parsing module to extract skills, experience, and education. Analyze skill relevance by considering experience duration and project context.	Introduces context-aware skill proficiency estimation, moving beyond keyword-based extraction to evaluate actual skill depth and relevance.
Deshan MAK IT22195616	Hiring Risk & Early Attrition Prediction	Develop an AI model to predict early resignation risk using employment history patterns, overqualification indicators, expectation-role mismatch, and interview insights. Generate attrition risk scores and retention probability.	Introduces predictive retention intelligence into recruitment systems, enabling organizations to avoid costly early attrition—an area largely absent in traditional ATS platforms.
Tennakoon TMMS IT22197832	AI-Based Interview Evaluation & Certification Validation	Design an AI-driven interview analysis module to assess communication and behavioral cues. Implement certification validation to verify authenticity, relevance, and validity of candidate credentials.	Combines soft-skill evaluation with automated certification validation, improving both candidate assessment accuracy and credential trustworthiness.

10. **This part is to be filled by the Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				